CBSE MOCK — Pre-Board Practice Series

Central Board of Secondary Education

CLASS X — MATHEMATICS

Date: 12 November 2026 | Maximum Marks: 80 | Time Allowed: 3 Hours

GENERAL INSTRUCTIONS:

1. This question paper contains 5 sections. Read the instructions for each section carefully.
2. All questions are compulsory unless stated otherwise.
3. Write all answers neatly and show relevant working wherever required.

(i) Calculators are NOT permitted.

4. Marks for each question are indicated against it.
5. This paper is a Practice Paper — it is similar to but NOT an actual CBSE board paper.
6. Download more FREE practice papers daily at: **www.cbsemock.org.in**

SECTION A — MCQ & Assertion-Reason (20 Marks)

A/R (1 mark each) | MCQ (1 mark each)

Q1. **[1 mark]**

If $(\tan\theta + \cot\theta) = 2$, then the value of $\tan^2\theta + \cot^2\theta$ is:

(A) 2
(B) 4
(C) 6
(D) 8

Q2. **[1 mark]**

The volume of a sphere is 4851 cm³. Its radius is: (Use $\pi = 22/7$)

(A) 7 cm
(B) 10.5 cm
(C) 14 cm
(D) 21 cm

Q3. **[1 mark]**

The coordinates of the mid-point of the line segment joining A(–3, 5) and B(7, –3) are:

(A) (2, 1)
(B) (–2, 1)
(C) (1, 2)
(D) (2, –1)

Questions 19 & 20 are Assertion-Reason type. Choose the correct option: (A) Both A and R are true and R is the correct explanation of A. (B) Both A and R are true but R is not the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true.

Q4. **[1 mark]**

Assertion (A): For $0 < \theta \leq 90°$, $\operatorname{cosec}\theta - \cot\theta$ and $\operatorname{cosec}\theta + \cot\theta$ are reciprocals of each other. Reason (R): $\operatorname{cosec}^2\theta - \cot^2\theta = 1$.

Q5. **[1 mark]**

The probability of getting a composite number when a die is thrown once is:

(A) 1/3
(B) 1/2
(C) 2/3
(D) 1/6

Q6. [1 mark]

The area of the sector of a circle of radius 6 cm and angle 60° is:

(A) 18.86 cm²
(B) 6π cm²
(C) 36π cm²
(D) 12π cm²

Q7. [1 mark]

Observe the graph of polynomial p(x). The number of zeroes of p(x) is:

(A) 5
(B) 4
(C) 6
(D) 3

Q8. [1 mark]

Assertion (A): The points (1, 7), (4, 2), (–1, –1) and (–4, 4) are the vertices of a square. Reason (R): If the diagonals of a quadrilateral are equal and bisect each other at right angles, then it is a square.

Q9. [1 mark]

If sin A = 3/5, then the value of (tan A + sec A) is:

(A) 2
(B) 1/2
(C) 3/2
(D) 5/3

Q10. [1 mark]

A bag has 5 red and 3 blue balls. If a ball is drawn at random, the probability that it is not blue is:

(A) 3/8
(B) 5/8
(C) 1/8
(D) 7/8

Q11. [1 mark]

If 3 sin θ = 4 cos θ, then sec θ is:

(A) 5/4
(B) 5/3
(C) 4/3
(D) 3/4

Q12. [1 mark]

The LCM of two numbers is 1200. Which of the following cannot be their HCF?

(A) 600
(B) 500
(C) 400
(D) 200

Q13. [1 mark]

If the point P(k, 0) divides the line segment joining A(2, –2) and B(–7, 4) in the ratio 1:2, then the value of k is:

(A) –1
(B) 1
(C) 2
(D) –2

Q14. [1 mark]

In an A.P., if the first term a = 7, common difference d = 3 and the last term l = 184, then the number of terms n is:

(A) 59
(B) 60
(C) 61
(D) 62

Q15. [1 mark]

It is given that ■ABC ~ ■EDF. Which of the following is not true?

(A) Perimeter of ■ABC/Perimeter of ■EDF = AB/ED
(B) AB/ED = AC/EF
(C) ∠A = ∠D, ∠C = ∠F
(D) (AB+BC)/AC = (DE+DF)/EF

Q16. [1 mark]

If the quadratic equation $9x^2 + 8kx + 16 = 0$ has real and equal roots, then the value of k is:

(A) 3
(B) –3
(C) –4
(D) 3/2

Q17. [1 mark]

A quadratic polynomial with sum of zeroes = 3 and product of zeroes = 2 is:

(A) $x^2 – 3x + 2$
(B) $x^2 + 3x + 2$
(C) $x^2 – 3x – 2$
(D) $x^2 + 3x – 2$

Q18. [1 mark]

The n■■ term of an A.P. is $\sqrt{2}\, n + 1$. Its common difference is:

(A) $\sqrt{2}$
(B) $\sqrt{2}\, n$
(C) 1
(D) $\sqrt{2} + 1$

Q19. [1 mark]

If the mean of the data: 7, 8, 9, 7, 8, 7, k is 8, then the value of k is:

(A) 10
(B) 11
(C) 12
(D) 13

Q20. [1 mark]

If PQ and PR are tangents to the circle with centre O and radius 4 cm such that ∠QPR = 90°, then the length OP is:

(A) 4 cm
(B) $4\sqrt{2}$ cm
(C) 8 cm
(D) $2\sqrt{2}$ cm

SECTION B — Very Short Answer (10 Marks)

VSA (2 marks each)

Q21. [2 marks]

The length of the minute hand of a clock is 14 cm. Find the area swept by the minute hand in 5 minutes. OR A chord of a circle of radius 10 cm subtends a right angle at the centre. Find the area of the minor segment. (Use $\pi = 3.14$)

Q22. [2 marks]

Find the value of $\sin 30° + \cos 60° - \tan 45° + \operatorname{cosec} 90°$.

Q23. [2 marks]

In the given figure, if TP and TQ are the two tangents to a circle with centre O so that $\angle POQ = 110°$, then find $\angle PTQ$.

Q24. [2 marks]

Solve the following pair of equations: $3x + 2y = 11$ and $2x + 3y = 4$. OR Find the value(s) of k for which the system of equations $x + 2y = 3$ and $5x + ky = 7$ has a unique solution.

Q25. [2 marks]

Find the HCF and LCM of 404 and 96 and verify that HCF × LCM = Product of the two numbers.

SECTION C — Short Answer (18 Marks)

SA (3 marks each)

Q26. [3 marks]

The sum of the first 20 terms of an AP is 400 and the sum of the first 40 terms is 1600. Find the sum of the first 60 terms. OR If 10■■ term of an AP is 52 and 17■■ term is 20 more than the 13■■ term, find the AP.

Q27. [3 marks]

Find the mean of the following data: Class: 0-10, 10-20, 20-30, 30-40, 40-50 Frequency: 7, 10, 15, 8, 10

Q28. [3 marks]

Find the roots of the quadratic equation: $x^2 - 3x - 10 = 0$ by the method of completing the square.

Q29. [3 marks]

Prove that $\sqrt{5}$ is irrational.

Q30. [3 marks]

Prove that the tangents drawn from an external point to a circle are equal in length.

Q31. [3 marks]

If $\sin\theta + \cos\theta = \sqrt{2}$, prove that $\tan\theta + \cot\theta = 2$. OR Prove that: $(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$.

SECTION D — Long Answer (20 Marks)

LA (5 marks each)

Q32. **[5 marks]**

A straight highway leads to the foot of a tower. A man standing at the top of the tower observes a car at an angle of depression of 30°. The car is approaching the foot of the tower with a speed of 50 m/min. After 6 minutes the angle of depression of the car is 60°. Find the height of the tower. OR The angle of elevation of the top of a building from the foot of the tower is 30° and the angle of elevation of the top of the tower from the foot of the building is 60°. If the tower is 50 m high, find the height of the building.

Q33. **[5 marks]**

State and prove the Basic Proportionality Theorem (Thales' Theorem). Using it, prove that the line joining the midpoints of two sides of a triangle is parallel to the third side and is half of it. OR Prove that in a right triangle, the square on the hypotenuse is equal to the sum of the squares on the other two sides. Using it, determine whether a triangle with sides 7 cm, 24 cm and 25 cm is a right triangle.

Q34. **[5 marks]**

A container shaped like a right circular cylinder having diameter 12 cm and height 15 cm is full of ice cream. The ice cream is to be filled into cones of height 12 cm and diameter 6 cm, having a hemispherical shape at the top. Find the number of cones which can be filled with ice cream. OR A heap of rice is in the form of a cone of base diameter 24 m and height 3.5 m. Find the volume of the rice. How much canvas cloth is required to just cover the heap?

Q35. **[5 marks]**

A train travels 360 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 1 hour less. Find the original speed of the train. OR A motor boat whose speed is 18 km/h in still water takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream.

SECTION E — Case Study (12 Marks)

CBQ (4 marks each — Case Study)

Q36. **[4 marks]**

A circular park of radius 20 m is situated in a colony. Three boys Ankur, Syed and David are sitting at equal distances on its boundary each having a toy telephone in their hands to talk each other. (a) Find the length of the string of each phone. (b) Find the area of the triangle formed by the three boys. (c) Find the distance of each boy from the centre of the park.

Q37. **[4 marks]**

A survey was conducted among Class X students about their daily screen time (in hours). The data collected is as follows: Screen time: 0-2, 2-4, 4-6, 6-8, 8-10; Frequency: 10, 15, 25, 30, 20. (a) Find the modal screen time. (b) Find the median screen time. (c) Find the mean screen time. OR (c) How many students have screen time less than 4 hours?

Q38. **[4 marks]**

India's National Parks are rich in biodiversity. A forest ranger is monitoring the spread of a particular plant species. The number of plants counted in different zones (in hundreds) is 3, 5, 7, 9, 11... (a) What type of sequence is this? (b) Find the number of plants in the 10■■ zone. (c) Find the total number of plants in the first 10 zones. OR (c) If a zone has 23 (hundred) plants, what is its zone number?

✦ END OF QUESTION PAPER ✦

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